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EXPERIMENT 1: Monthly Sales Analysis

Step 1: Load

Package

```
library(ggplot2) library(plotly)
```

Step 2: Create Dataset sales <-

```
data.frame(      Month      =  
c("Jan","Feb","Mar","Apr","May"),  
  Sales = c(1200,1500,1800,1600,2000)  
)
```

```
print(sales) Step
```

3: Line Chart

```
ggplot(sales, aes(x = Month, y = Sales, group =  
  1)) + geom_line(color = "blue", linewidth = 1.2)  
+ geom_point(color = "red", size = 3) + labs(  
  title = "Monthly Sales  
  Trend", x = "Month", y  
  = "Sales"  
) +
```

```
theme_minimal() Step
```

4: Bar Chart

```
ggplot(sales, aes(x = Month, y = Sales, fill = Month)) +  
  geom_bar(stat = "identity") + labs(  
    title = "Monthly Sales  
    Comparison", x = "Month", y  
    = "Sales"  
  ) +
```

```
theme_minimal() Step
```

5: Scatter Plot

```

ggplot(sales, aes(x = Month, y = Sales))
+ geom_point(size = 4, color = "blue")
+ geom_smooth(aes(group = 1), method = "lm", se = FALSE, color
=
"red") + labs( title = "Monthly
Sales Scatter
Plot", x = "Month", y
= "Sales"
) +

```

```

theme_minimal() Step 6:
Interactive Dashboard p1 <-
plot_ly(sales, x =
~Month, y
= ~Sales, type =
"scatter",
mode = "lines+markers")

```

```

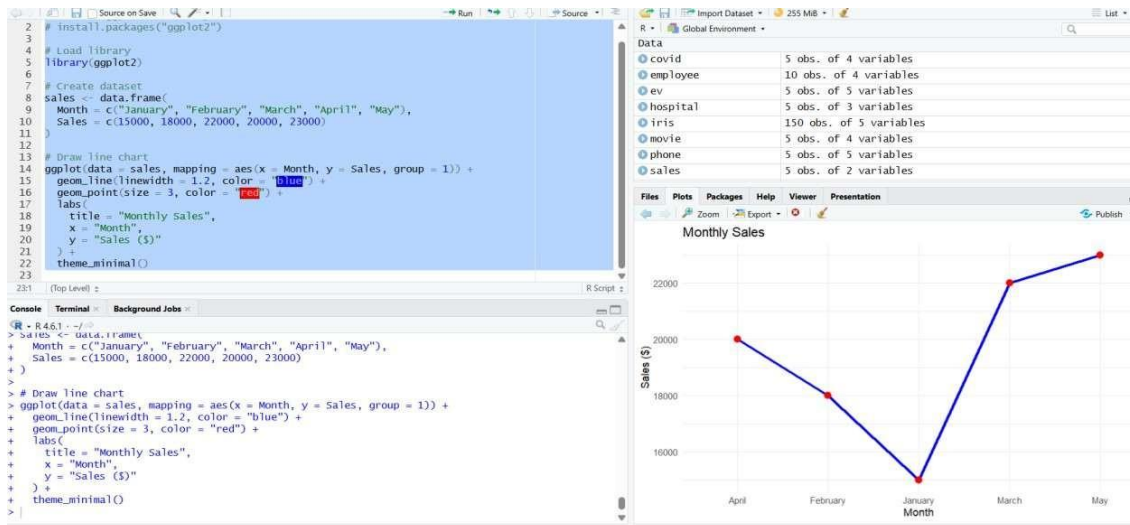
p2 <- plot_ly(sales, x
=
~Month, y
= ~Sales,
type =
"bar")

```

```

subplot(p1, p2, nrow = 2)

```



EXPERIMENT 2: Customer Feedback Analysis

PROGRAM

Step 1: Load Packages

```
library(ggplot2)
```

```
library(wordcloud)
```

```
library(RColorBrewer)
```

Step 2: Create Dataset

```
customer <- data.frame(
  Age = c(20,25,30,35,40),
  Satisfaction = c("High","Medium","High","Low","Medium")
)
```

```
print(customer) Step 3:
```

```
Histogram ggplot(customer,
aes(x = Age)) +
geom_histogram(binwidth =
5, fill = "skyblue",
color = "black") + labs(
```

```

title = "Customer Age
Distribution", x = "Age",
y = "Frequency"
) +
theme_minima

```

l) Step 4: Pie

```

Chart pie( table(customer$Satisfacti
on), col = rainbow(3), main =
"Customer Satisfaction Level"
)

```

Step 5: Stacked Bar Chart

```

customer$Age_Group <- c("20-25","20-25","30-35","30-35","40+")
ggplot(customer,
aes(x =
Age_Group, fill
= Satisfaction)) +
geom_bar() + labs( title =
"Satisfaction by Age Group", x =
"Age Group", y = "Count"
) +
theme_minimal()

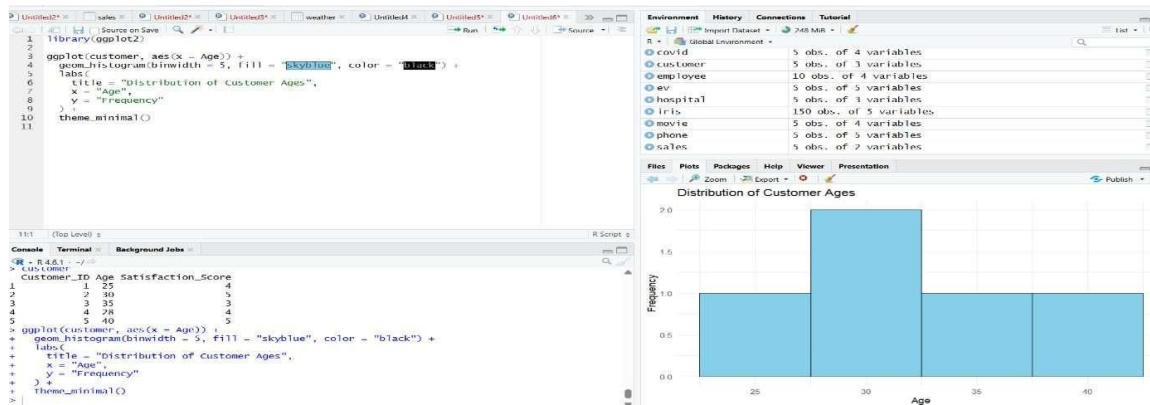
```

Step 6: Word

```

Cloud wordcloud( words =
customer$Satisfaction, freq =
c(2,2,1,1,2), colors =
brewer.pal(8,"Dark2")
)

```



EXPERIMENT 3: Employee Performance Analysis

PROGRAM

Step 1: Load Packages

```
library(ggplot2)
```

library(plotly) Step 2:

Create Dataset employee

```
<- data.frame(
```

```
Employee_ID =
```

```
c(1,2,3,4,5),
```

```
Department = c("Sales","HR","Marketing","Sales","HR"),
```

```
Years_of_Service = c(5,3,7,4,2),
```

```
Performance_Score = c(85,92,78,90,76)
```

```
)
```

print(employee) Step

3: Line Chart

```
ggplot(employee,
```

```
  aes(x =
```

```
    Years_of_Service, y =
```

```
    =
```

```
    Performance_Score,
```

```
    group = 1)) + geom_line(color =
```

```
"blue", linewidth =
```

```

1.2) + geom_point(color = "red", size = 3)
+ labs(
  title = "Employee Performance
Trend", x = "Years of Service",
y = "Performance Score"
) +
theme_minima
l() Step 4: Bar
Chart
ggplot(employe
e,
  aes(x = Department,
fill =
Department)) + geom_bar()
+
labs( title = "Employees by
Department", x =
"Department", y = "Number
of Employees"
) +
theme_minimal()
Step 5: Scatter Plot
ggplot(employee, aes(x =
Years_of_Service,
y = Performance_Score)) + geom_point(size
= 4, color = "blue") +
geom_smooth(method = "lm",
se = FALSE, color =
"red") +
labs(
title = "Years of Service vs

```

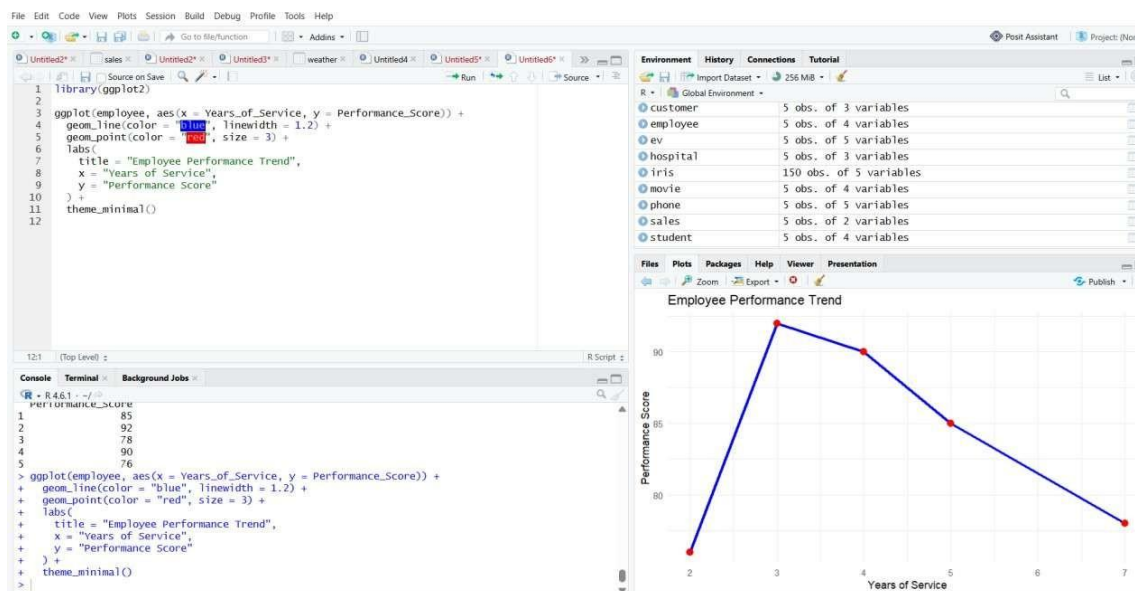
```

Performance", x = "Years of
Service",      y      =
"Performance Score"
) +
theme_minimal() Step 6:
Interactive Dashboard p1 <-
plot_ly(employee,      x      =
~Years_of_Service, y =
~Performance_Score,
type = "scatter",
mode = "lines+markers")

p2 <- plot_ly(employee,
x = ~Department,
type = "bar")

subplot(p1, p2, nrow = 2)

```



EXPERIMENT 4: Product Inventory Analysis

PROGRAM

Step 1: Load Packages

```
library(ggplot2) library(plotly)
```

Step 2: Create Dataset inventory

```
<- data.frame(
  Product_Name = c("Product A","Product B","Product C","Product D","Product
E"),
  Category =
c("Electronics","Electronics","Furniture","Furniture","Furniture"),
  Price =
c(500,650,450,700,550), Quantity = c(250,175,300,200,220)
)
```

```
print(inventory)
```

Step 3: Bar Chart

```
ggplot(inventory,
aes(x =
  Product_Name, y
  = Quantity,
  fill =
Product_Name)) +
geom_bar(stat =
"identity") + labs(
  title = "Product
Inventory", x
  = "Product Name", y
  = "Quantity"
) +
```

theme_minimal() Step 4:

Stacked Bar Chart

```
ggplot(inventory, aes(x =
  Category, y =
  Quantity,
  fill =
Product_Name)) +
geom_bar(stat =
"identity") + labs(
```



```

title = "Inventory by
Category", x = "Category", y
= "Quantity"
) +
theme_minimal()

```

Step 5: Scatter Plot ggplot(inventory

```

, aes(x =
Price, y = Quantity)) +
geom_point(size = 4, color =
"blue") + geom_smooth(method
= "lm", se = FALSE,
color = "red") +
labs(
title = "Price vs
Quantity", x = "Price",
y = "Quantity"
) +

```

theme_minimal() Step 6:

Interactive Dashboard p1 <-

```

plot_ly(inventory, x =
~Product_Name,
y = ~Quantity,
type = "bar")

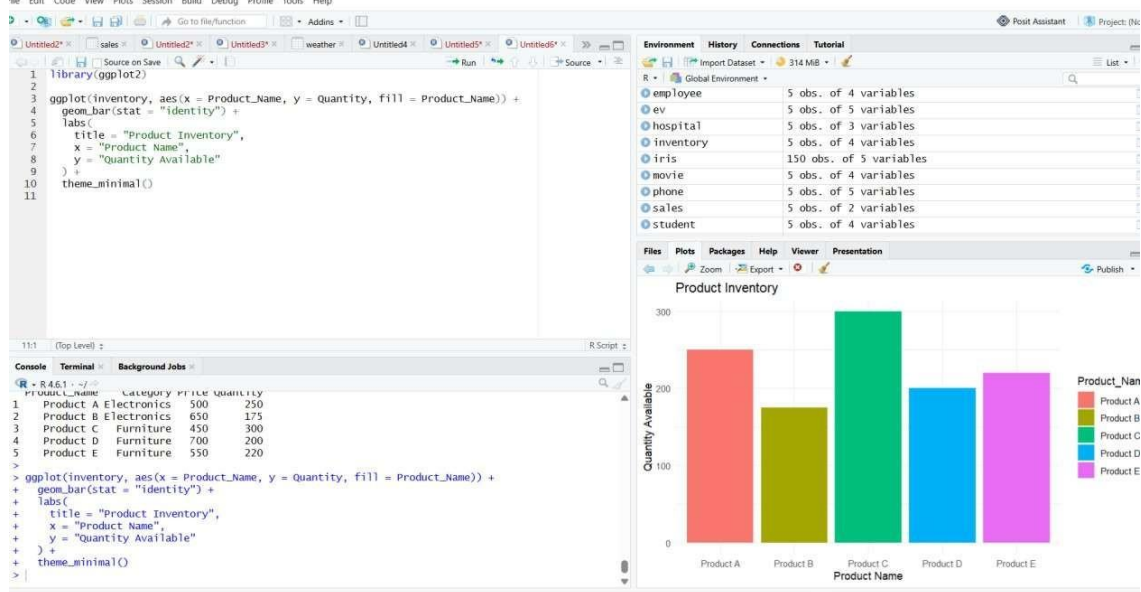
```

```

p2 <- plot_ly(inventory,
x = ~Price, y =
~Quantity, type
= "scatter",
mode =
"markers")

```

subplot(p1, p2, nrow = 2)



SET 5: Website

PROGRAM

Step 1: Load Packages

```
library(ggplot2)
```

```
library(plotly)
```

```
library(tidyr)
```

Step 2: Create Dataset traffic

```
<- data.frame(
```

```
Date = as.Date(c(
```

```
"2023-01-01",
```

```
"2023-01-02",
```

```
"2023-01-03",
```

```
"2023-01-04",
```

```
"2023-01-05"
```

```
)),
```

```
Page_Views =  
c( 1500, 1600,  
1400,  
1650,  
1800  
),
```

```
CTR = c(  
2.3,  
2.7,  
2.0,  
2.4, 2.6  
)
```

```
)
```

```
print(traffic)
```

Question 1 Line Chart

```
ggplot(traffic,  
aes(x = Date,  
y =  
Page_Views,  
group = 1)) + geom_line(color =  
"blue", linewidth = 1.2)  
+
```

```
geom_point(color =  
"red", size = 3) +
```

```

labs( title = "Daily Website
Page Views", x = "Date", y
= "Page Views"
) +

theme_minimal()

```

